

CS-1180 Lab Problem 6: Multiply and Verify

PURPOSE: To test the ability to write methods for modular code

DIRECTIONS: Write a method `printTable` that takes a number `n` as a parameter and prints a formatted multiplication table from 1 to `n`.

Write a method `verifyNum` that takes a number `n` and returns a `boolean` result after checking if `n` is greater than 0 but less than 50.

Test the methods by calling them from `main`. The `main` method should prompt the user for the value of `n` and use the return value of the `verifyNum` method. If `verifyNum` returns `true`, call the `printTable` method. If `verifyNum` returns `false`, print `Error: n must be between 0 and 50` and exit the program.

Use JavaDoc standards for your methods - a block comment with a minimum of:

- method description
- `@param` tag(s)
- `@return` tag(s)

EXAMPLE: Sample JavaDoc block comment:

```
/**
 * Method description
 * @param talk about int a
 * @return what value is returned
 */
public static int demo (int a){
    // code here
}
```

Sample program execution:

```
What is the value of n?
> 6
1  2  3  4  5  6
2  4  6  8  10 12
3  6  9  12 15 18
4  8  12 16 20 24
5  10 15 20 25 30
6  12 18 24 30 36

What is the value of n?
> -4
Error: n must be between 0 and 50
```

RUBRIC:

(3 pts) Base Functionality

- [1] Reads input correctly
- [1] Output is correct
- [1] Handles negative input correctly

(1 pts) Style

- [1] The code is meaningfully commented

IMPORTANT NOTES:

- **If your submission doesn't utilize methods or doesn't have proper javadocs, you will receive a zero**
- Submissions that do not compile will receive a zero
- Submissions with improperly cited AI will receive a zero and an academic integrity violation
- Submissions that are partially or fully copied from another submission will receive a zero and an academic integrity violation